

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			Indicative content PHYSICAL STATE 1. carbon dioxide and carbon monoxide have weak forces between separate molecules (so it is a gas) 2. lead has a relatively low ionisation energy so can form positive ions; its oxides are ionic with strong attraction between ions (so it they are solids) ACID/BASE 3. carbon dioxide is a non-metal / covalent oxide so it is an acidic oxide 4. lead(II) oxide reacts with both acids and bases so it is an amphoteric oxide REDOX 5. stability of +2 oxidation state increases down the group / +4 most stable for top of group and +2 most stable at bottom of group / inert pair effect increases down the group 6. carbon monoxide contains carbon in oxidation state +2 whilst +4 is stable, so carbon will be easily oxidised / so it acts as a reducing agent 7. lead(IV) oxide contains lead in oxidation state +4 whilst +2 is stable, so lead will be easily reduced / so it acts as an oxidising agent Relevant equations: • CO as a reducing agent e.g. $\text{CO} + \text{CuO} \rightarrow \text{Cu} + \text{CO}_2$ • PbO_2 as an oxidising agent e.g. $\text{PbO}_2 + 4\text{HCl} \rightarrow \text{PbCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$ • CO_2 reacting with a base: $\text{CO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$ or $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}^+ + \text{HCO}_3^-$ • PbO reacting with a base: $\text{PbO} + 2\text{NaOH} \rightarrow \text{Na}_2\text{PbO}_2 + \text{H}_2\text{O}$ • PbO reacting with an acid: $\text{PbO} + 2\text{HNO}_3 \rightarrow \text{Pb}(\text{NO}_3)_2 + \text{H}_2\text{O}$	3	3		6		2